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Simulation of BLDC Motor Control Using Conventional PI Controller in MATLAB Simulink



Abstract: - This research Paper presents a comprehensive analysis of Brushless DC (BLDC) motors, focusing on their construction, operating principles, and control methodologies, particularly in variable-speed applications. BLDC motors are highlighted for their advantages over traditional DC brushed and AC induction motors, including enhanced efficiency, longer operational lifespan, and suitability for environments sensitive to electromagnetic interference. The paper details the essential components of a modern variable-speed drive system and emphasize the role of Hall effect sensors in rotor position sensing, which is critical for effective motor control. A simulation model employing a PI controller for closed-loop operation of a BLDC motor is developed, demonstrating the implementation of a Hysteresis Current Controller to maintain current within a specified band. The study encapsulates the relationship between speed and back EMF in BLDC motors and showcases the effectiveness of the proposed control strategy through simulation results. Overall, the findings underscore the potential of BLDC motors in high-performance applications, with implications for future research in advanced motor control techniques.

Keywords: Brushless Direct Current Motor (BLDC), alternating current (AC), direct current (DC), Proportional Integral (PI), Proportional Integral Derivative (PID), Electromotive Force (EMF).

I INTRODUCTION

To design a high-performance drive system, the engineer needs a deep understanding of machine performance, dynamic modeling, and parameter variation. Industrial drive applications are generally divided into two categories: constant-speed and variable-speed drives. Drive. Formerly for constant-speed applications Alternating current (AC) motors are powered by constant-frequency sinusoidal supplies, while direct current (DC) motors types and variable speeds are preferred.

But DC brush machines have many drawbacks, such as high cost, high rotor inertia, and maintenance difficulties associated with commutator brushes, commutator and brushes can limit machine speed and peak current, causing electromagnetic interference (EMI) problems have occurred, as well as using the device in dirty or explosive environments to make unsuitable. On the other hand DC mechanical drive converters and control systems are straight forward; this mechanical torque response is very fast. In contrast, AC motors have non-linear kinetic torque characteristics that vanish, resulting in reduced torque and increased rotor inertia at low speeds.

Brushless DC motors are often used where precise speed control is required, CDs in computer disk drives or video cassette recorders, spindles in CD-ROM (etc.) drives, and office supplies as fans, laser printers and photocopiers inside than conventional motors. In have several advantages: DC brushless motor without a transformer to charge life can be much longer compared to DC motor brushes and transformers are used about. The changes also tend to cause significant increases in electrical and RF noise; it does not have a commutator or brushes.

A modern variable-speed system has four components:

- 1) Electric machine – ac or dc.
- 2) Power Converter – rectifier, chopper, inverter and cycloconverters.

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3) Controller- Matching the motor and power converter to meet load requirements. Such as PI, PID- controller, Sliding Mode Controller, Fuzzy logic controller etc.

4) Load

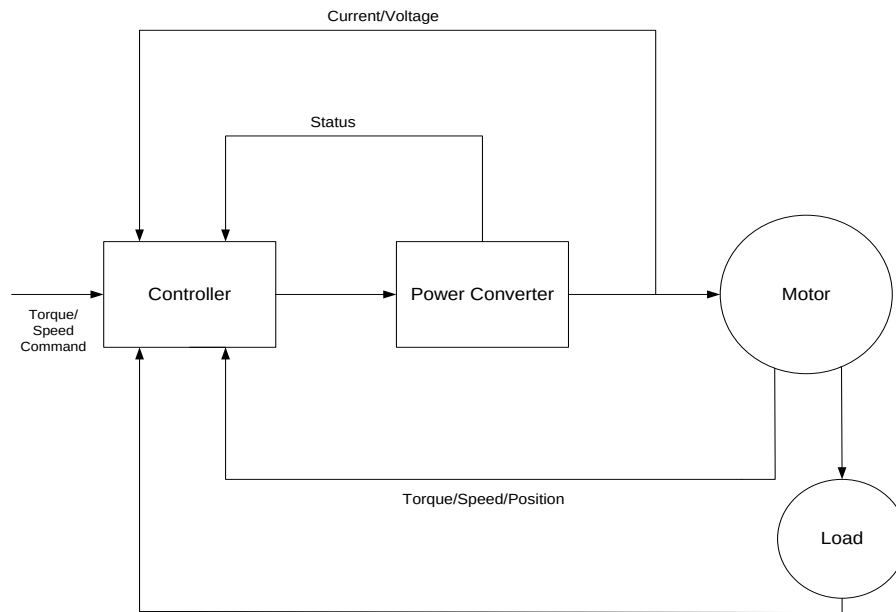


Figure-1 motor drive schematic

BLDC motors have many advantages over brushed DC motors and induction motors. A few of these are:

Better speed versus torque characteristics

- High dynamic response
- High efficiency
- Long operating life
- Noiseless operation
- Higher speed ranges [5]

II. CONSTRUCTION AND OPERATING PRINCIPLE

Construction

BLDC motors are a type of synchronous motor. This means the magnetic field generated by the stator and the magnetic field generated by the rotor rotates at the same frequency. BLDC motors do not experience the slip that is normally seen in induction motors. BLDC motors come in single-phase, 2-phase and 3-phase configurations. Corresponding to its type, the stator has the same number of windings. Out of these, 3-phase motors are the most popular and widely used. [5]

Stator

The stator of a BLDC motor consists of stacked steel laminations with windings placed in the slots that are axially cut along the inner periphery as shown in below Figure.2 Traditionally, the stator resembles that of an induction motor; however, the windings are distributed in a different manner.

Most BLDC motors have three stator windings connected in star fashion. Each of these windings are constructed with numerous coils interconnected to form a winding. One or more coils are placed in the slots and they are interconnected to make a winding. Each of these windings are distributed over the stator periphery to form an even numbers of poles. There are two types of stator windings variants: trapezoidal and sinusoidal motors. This differentiation is made on the basis of the interconnection of coils in the stator windings to give the different types of back Electromotive Force (EMF).

Rotor

The rotor is made of permanent magnet and can vary from two to eight pole pairs with alternate North (N) and South (S) poles. Based on the required magnetic field density in the rotor, the proper magnetic material is chosen to make the rotor. Ferrite magnets are traditionally used to make permanent magnets. As the technology advances, rare earth alloy magnets are gaining popularity. The ferrite magnets are less expensive but they have the disadvantage of low flux density for a given volume. In contrast, the alloy material has high magnetic density per volume and enables the rotor to compress further for the same torque. Also, these alloy magnets improve the size-to-weight ratio and give higher torque for the same size motor using ferrite magnets.

Hall sensor

A Hall Effect sensor is a transducer that varies its output voltage in response to changes in magnetic field. Hall sensors are used for proximity switching, positioning, speed detection, and current sensing applications.

In its simplest form, the sensor operates as an analogue transducer, directly returning a voltage. With a known magnetic field, its distance from the Hall plate can be determined. Using groups of sensors, the relative position of the magnet can be deduced.

Electricity carried through a conductor will produce a magnetic field that varies with current, and a Hall sensor can be used to measure the current without interrupting the circuit. Typically, the sensor is integrated with a wound core or permanent magnet that surrounds the conductor to be measured.

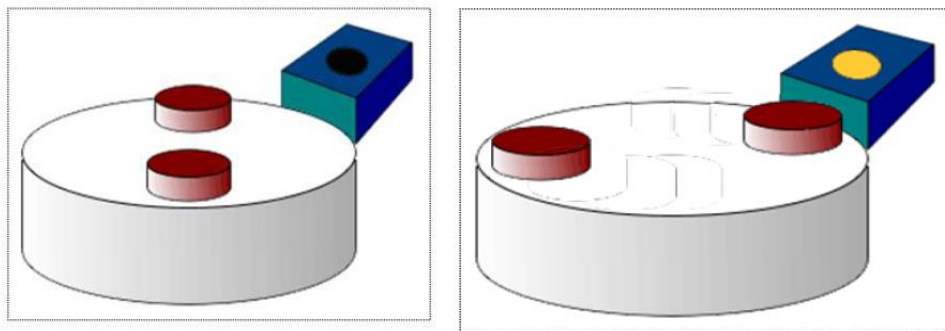


Figure-2 Principle of Hall Sensor

Frequently, a Hall sensor is combined with circuitry that allows the device to act in a digital (on/off) mode, and may be called a switch in this configuration. Commonly seen in industrial applications such as the pictured pneumatic cylinder, they are also used in consumer equipment; for example some computer printers use them to detect missing paper and open covers. When high reliability is required, they are used in keyboards

Hall sensors are commonly used to time the speed of wheels and shafts, such as for internal combustion engine ignition timing, tachometers and anti-lock braking systems. They are used in brushless DC electric motors to detect the position of the permanent magnet. In the pictured wheel with two equally spaced magnets, the voltage from the sensor will peak twice for each revolution. This arrangement is commonly used to regulate the speed of disc drives.

Working of Hall Sensors

To rotate the BLDC motor the stator windings should be energized in a sequence. It is important to know the rotor position in order to understand which winding will be energized following the energizing sequence. Rotor position is sensed using Hall-effect sensors embedded into the stator. Most BLDC motors have three Hall sensors inside the stator on the non-driving end of the motor. Whenever the rotor magnetic poles pass near the Hall sensors they give a high or low signal indicating the N or S pole is passing near the sensors. Based on the combination of these three Hall sensor signals, the exact sequence of commutation can be determined.

Figure 3 shows a transverse section of a BLDC motor with a rotor that has alternate N and S permanent magnets. Hall sensors are embedded into the stationary part of the motor.

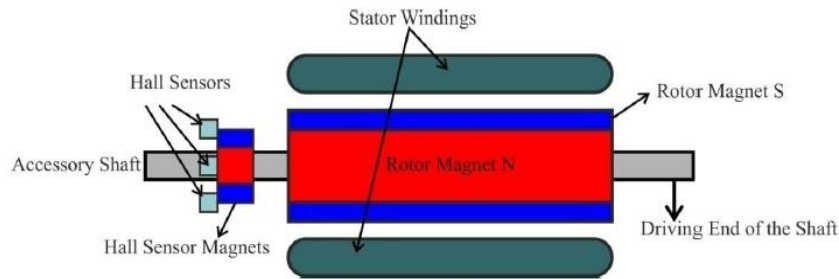


Figure-3 construction of brushless dc motor

The Hall sensors are normally mounted on a printed circuit board and fixed to the enclosure cap on the non-driving end. This enables users to adjust the complete assembly of Hall sensors to align with the rotor magnets in order to achieve the best performance. Based on the physical position of the Hall sensors, there are two versions of output. The Hall sensors may be at 60° or 120° phase shift to each other. Based on this, the motor manufacturer defines the commutation sequence, which should be followed when controlling the motor. Hall sensor uses to sense the position of rotor of BLDC motor and that signals are given to inverter circuit for controlling purpose.

Operation of BLDC Motor

BLDC works on basic principal of motor, when current carrying conductor placed in magnetic field it experiences force and its direction is define by Fleming left hand rule and force is define by $F=BIL$.

In BLDC, stator has winding but rotor has no winding so it does not require brushes. Rotor has permanent magnet in cylindrical form which continuously magnetic field generating. BLDC motor requires controlled alternating cycle voltages which are generated by Inverter. The main function of inverter is converts fixed direct cycle voltage in to variable alternating cycle voltage.

PWM means pulse width modulation technique is used to operate inverter and reduces harmonic contents. Pulse width is changing but amplitude of pulse is not changing. PWM is one of type of PWM techniques among them PWM is widely used in industries for controlling purpose.

The three phase BLDC Motor is operated in a two-phase-on fashion, i.e. here third phase is off, so two phases produce the highest torque, which two phases are energized depends on the rotor position. The signal generated from the position sensors in the form of three digit number that change over every 60 degree (electrical deg.). The figure shows also ideal current and BACK EMF waveforms.

Each interval starts with the rotor and stator field lines 120 degree apart and ends when they are 60 degree apart. Maximum torque is reached when the field lines are perpendicular: ^[1].

Equivalent diagram of BLDC motor

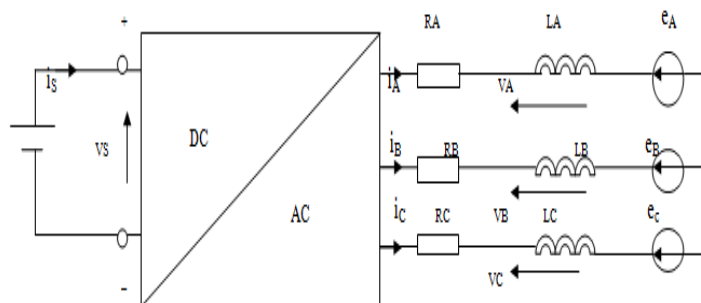


Figure-4 Equivalent diagram of BLDC motor

III. BLDC MOTOR CONTROL

Speed and current control loop for 180 degree commutation

For the characteristics of BLDCM, back-emf is directly proportional to the motor speed, and the torque production is almost directly proportional to phase current [5]. Because of these factors, the following drive scheme is commonly used in speed and current control, which is shown in figure-8. The implementation of this 180-degree commutation is challenging due to more complicated operation of reference current generated and the stability of speed control. The actual speed information is obtained from the feedback position information.

By comparing the difference between the actual speed and the reference speed, speed error is calculated. PI controller is implemented to generate the reference current according to the logic of the proposed 180-degree commutation current controller. In the Hysteresis current regulator, the power transistors are switched on or off depend on whether the current is larger or less than the generated 180-degree commutation current. This scheme works properly with excellent performance results for 180-degree commutation system.

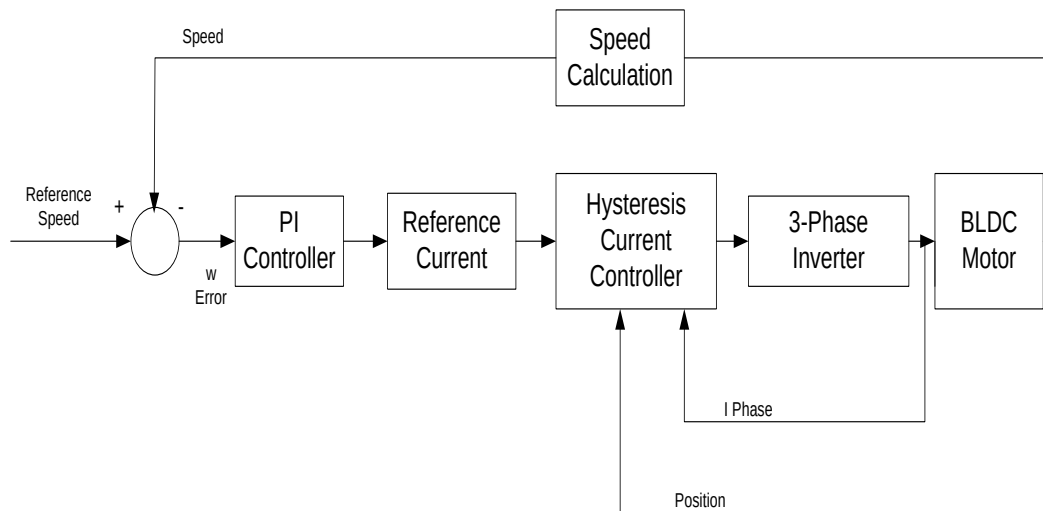


Figure-5 Speed and Current Control Loop of BLDC

Hysteresis current controller

Hysteresis current controller can also be implemented to control the inverter currents. The controller will generate the reference currents with the inverter within a range which is fixed by the width of the band gap. In this controller the desired current of a given phase is summed with the negative of the measured current. The error is fed to a comparator having a hysteresis band. When the error crosses the lower limit of the hysteresis band, the upper switch of the inverter leg is turned on. But when the current attempts to become less than the upper reference band, the bottom switch is turned on. The hysteresis band with the actual current and the resulting gate signals. This controller does not have a specific switching frequency and changes continuously but it is related with the band width [4].

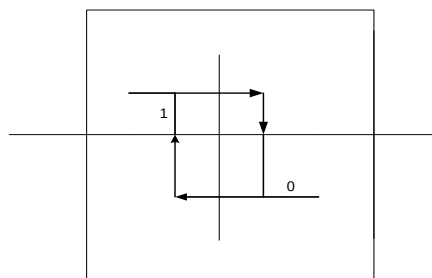


Figure-6 Typical hysteresis current controller

In hysteresis-band current control the actual current tracks the form of command current within a hysteresis band. In this approach the reference current wave is compared to the actual current wave thus producing the current error. When the current error exceeds a predefined hysteresis band, the upper switch in the half-bridge is turned off and the lower switch is turned on. As the current error goes below the hysteresis band, the opposite switching takes place. The principle of hysteresis band current control is illustrated in Figure. Peak to peak current ripple and switching frequency are related to the width of hysteresis band.

Hysteresis-band current control is very popular because it is simple to implement, has fast transient response, direct limiting of device peak current and practical insensitivity to machine parameters because of the elimination of any additional current controllers. However, PWM frequency is not fixed which results in non-optimal harmonic ripple in machine current. The current error is not strictly limited and may leave the hysteresis band. There is no interaction between the three phases. This drawback limits the application of this approach into special applications [4].

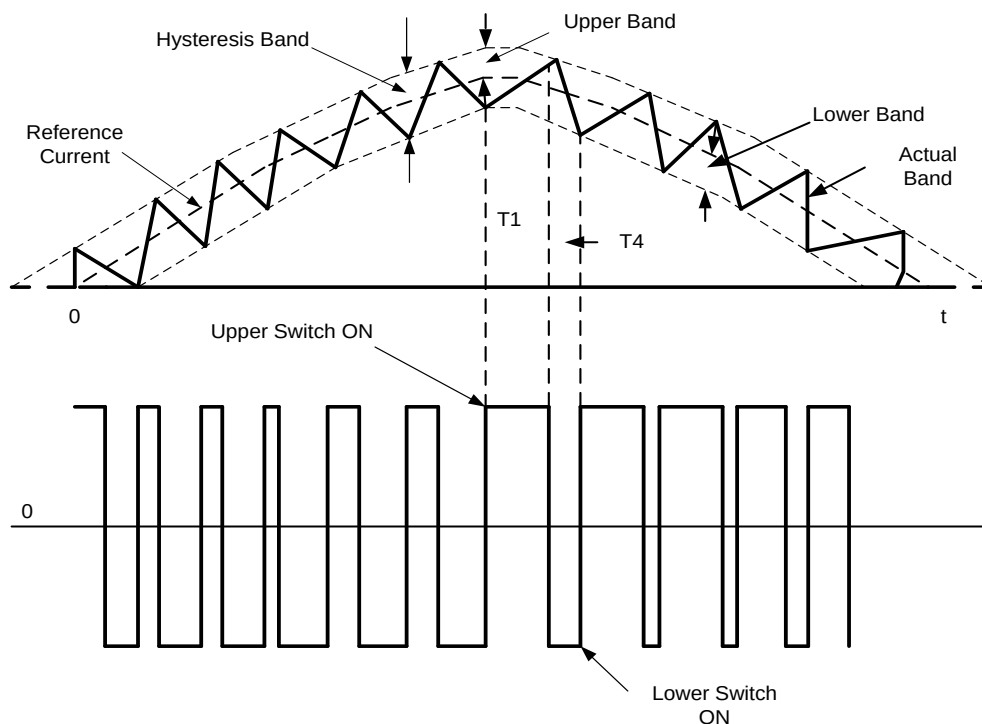


Figure-7 Principle of Hysteresis-band Current Control

IV. Simulation Model and Results with Conventional PI Controller

In this Model the motor parameters are put in to the state-space model and that state space output is the Stator Current of three phases, speed and theta. Here maxon motor data is taken from maxon motor website. The output of the speed is feedback into the reference speed and that two inputs are given to the sum block. This speed error is generated and given to the PI Controller input. This PI Controller Output is given to the Reference Current and that signal pass in the Hysteresis Current Controller. Then three pulses are generated which is input of inverter.

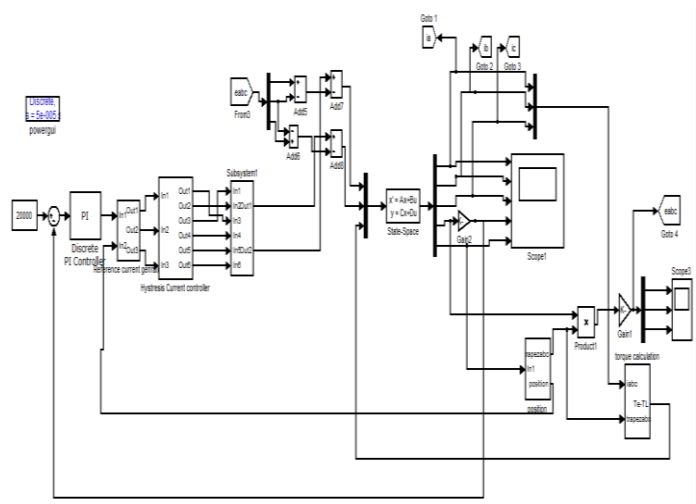


Figure-8 Simulation using Conventional Controlling

Stator Current Output Waveforms

Stator Current of Phase-A, Phase-B, and Phase-C are shown below which are the output of the State Space. Current passing through each Phase and the sinusoidal waveforms as shown below.

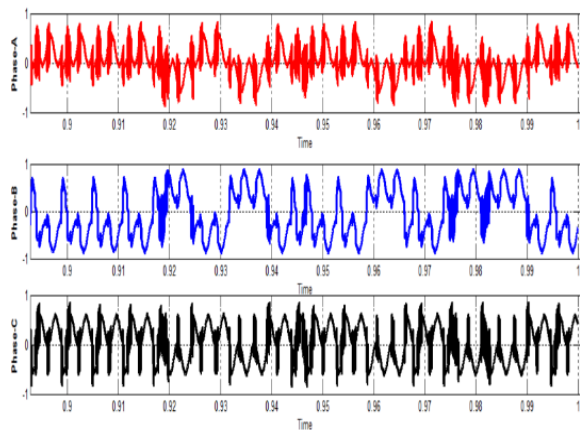


Figure- 9 Stator current waveform

Model of trapezoidal wave and Rotor Position

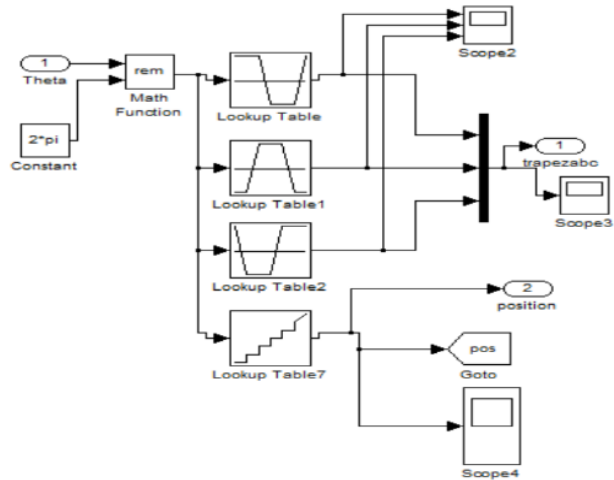


Figure- 10 Trapezoidal Wave and Rotor Position

Here trapezoidal wave and Position Calculation model Shown. These equations are implementing in above model where Look-up tables are implemented to obtain trapezoidal windings. The first three tables are implemented for a trapezoidal windings and one table for the rotor position. The rotor position is call back to the reference current calculations generation. The below figures shows the Trapezoidal wave output waveforms for the phase-A, phase-B, phase-C respectively. The rotor position is divided in to six sectors as shown in model.

Trapezoidal Wave Output

The Output Waveform shown below. Here, Trapezoidal waves are divided into 120 degree apart for each phase. Here shown below waveforms are Zoomed views of Trapezoidal waves.

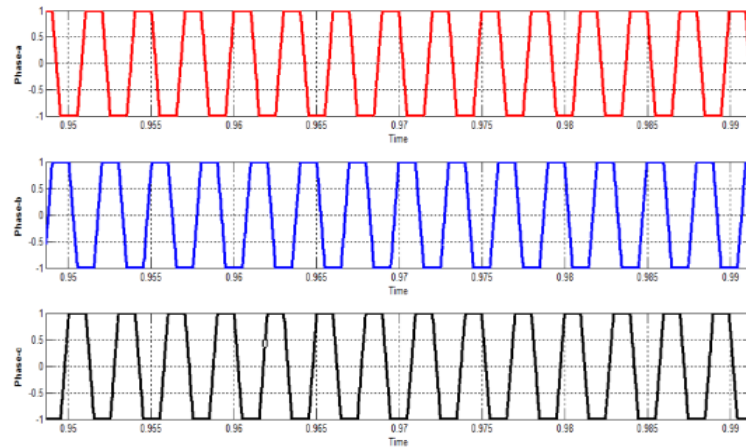


Figure- 11 Trapezoidal Waveforms

Torque Calculation

The torque calculation is the Dot Product of Trapezoidal waveforms and three Stator Currents. The output is multiplied by the gain, resulting in the electrical torque T_e .

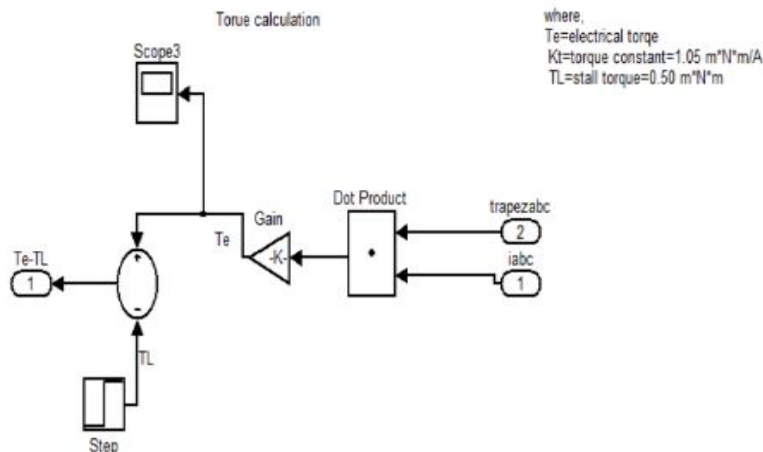


Figure- 12 Torque Calculation

Electrical torque

The Electrical Torque is as shown in fig which is dot product of trapezoidal wave and current. Here motor is small so that produced electric torque is also small. Almost sinusoidal torque is coming which shown in fig.

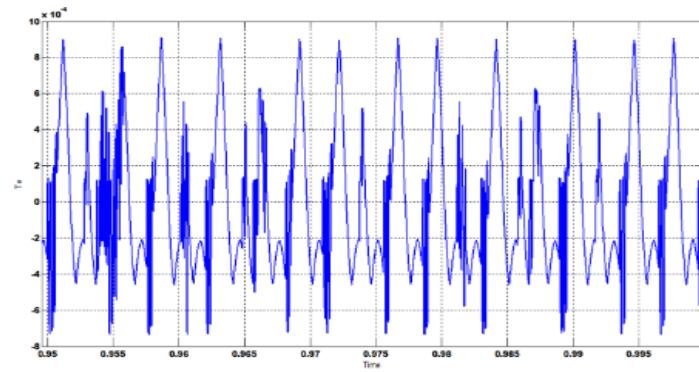


Figure-13 Electrical torque Output

Back-EMF Calculation

The Back-EMF Calculation is the product of speed and trapezoidal wave forms. That product is multiply with constant gain and the Back EMF desired Waveforms are for the phase-a, phase-b and phase-c respectively. The below figure Shows that the simulation model of a Back-EMF Calculation and the output waveforms.

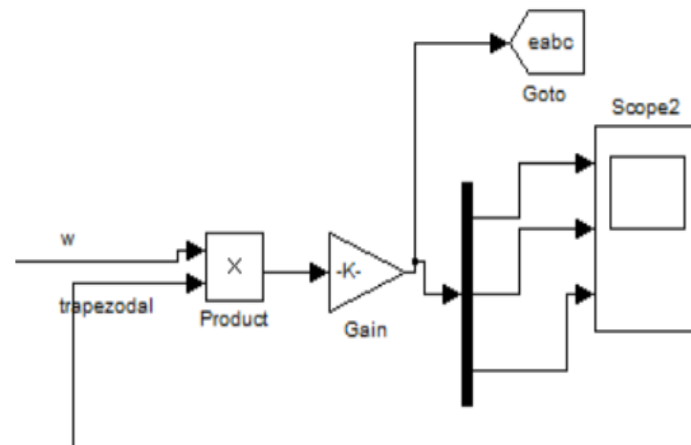


Figure- 14 Back EMF Calculation

Back –EMF Wave Forms

Below output waveforms of the Back-Emf of Sensor-less BLDC motor. The Back Emf is the Product of Trapezoidal wave and the Speed. Here we Considering trapezoidal winding so that these trapezoidal waves shown below.

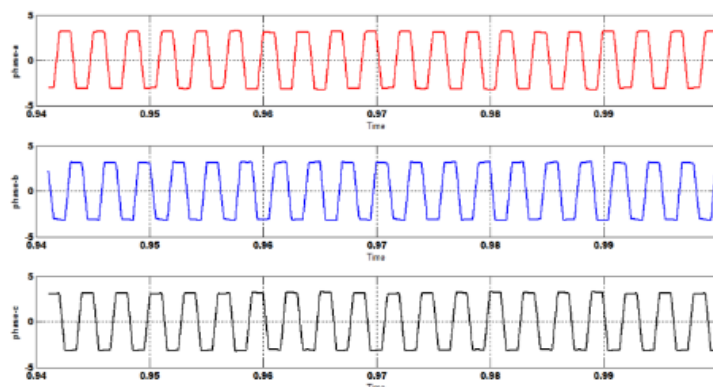


Figure- 15 Output Waveforms of Back-EMF

Hysteresis Current Controller

The Hysteresis Current Controller is the Controller which Control the Current within the Band which decided. In a hysteresis Current Controller there is upper limit and the lower limit of band is decided and that within that band the Controller Work. Three Outputs are Connected with NOT Gate and Total Six Pulses are Generated. These Six Pulses are the input of the Inverter. The Hysteresis Current Controller Model and Output Waveforms are shown below.

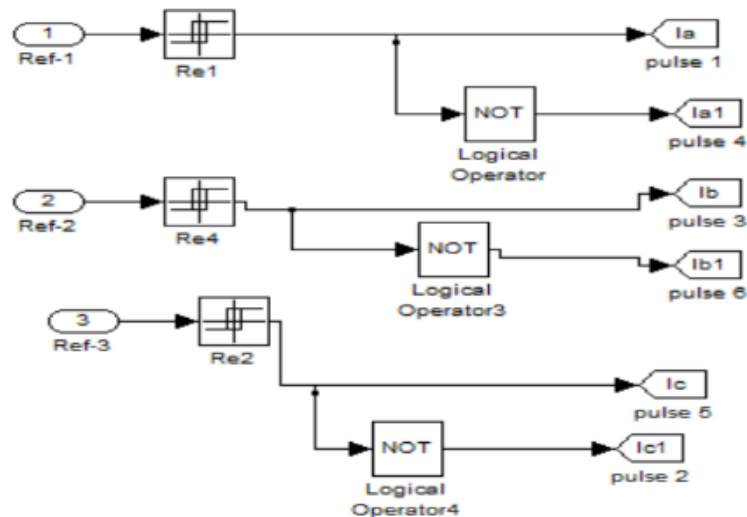


Figure- 16 Hysteresis Current Controller

Inverter Model

Inverter Model is shown below. The Ideal Switch is using as a Power electronics Device. Diode is connected to Anti-parallel to these devices. The pulses are given to Inverter as the Switching of Upper and lower Switch as shown in model.

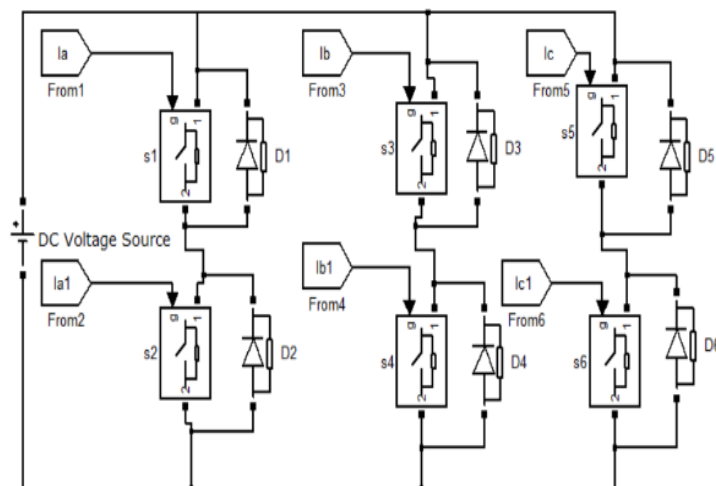


Figure-19 Inverter Model

Speed of BLDC Motor

Speed response is shown below using the Hysteresis current controller and using the Conventional PI Controller. By the shown output waveforms it is say that here some ripples are generated using this method.

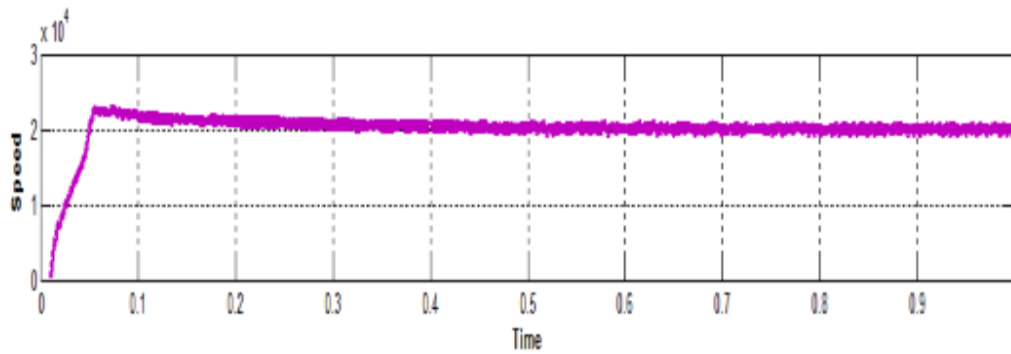


Figure- 17 Speed response of BLDC motor

V. CONCLUSION

Simulation results in the this paper, a BLDC motor with three-phase inverter has been simulated in 120 degree conduction mode using a Hall sensor. The PI controller is designed and simulated for closed-loop operation of the BLDC motor. BLDC motor is high speed, linear torque-speed relationship, and brushless motor. There are so many different methods for controlling of BLDC Motor like microprocessor based, with sensor or without sensor based and proportional integral based, for controlling BLDC Motor. Speed and position control can implemented using three control strategies like Variable D.C. Link Voltage control, PWM control and Hysteresis Band control. Here Hysteresis Band Controlling Method is used the Control the Motor. From the results, as speed increases, Back EMF will increase. The Hysteresis Band control method is easy to implement. Using the PI Controller the Output of the speed is achieved.

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